AI Club

Workshop #3

It's All About

Deep Learning for ECG Analysis

In the part of ACSE HACKATHON 2026

Check Attendance
&&
Discord QR Code

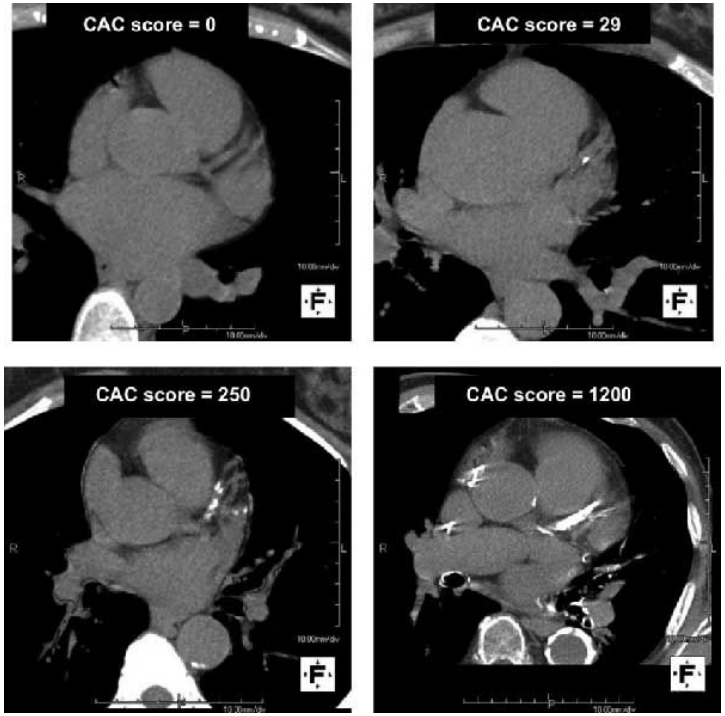
Agenda 💗

- **Intro to ECG and Coronary Artery Calcium Score**
- **Deep learning for ECG Analysis**
  - Model Training Pipeline
- **Hand-on Workshop**
- **Info Session for Hackathon**

Why Coronary Artery Calcification (CAC) Matters? 💗

Coronary Artery Calcification (CAC) refers to hardened calcium deposits that build up in the walls of coronary arteries; a major risk factor for heart attacks. Currently, CAC is detected using CT Scan CAC Scoring.

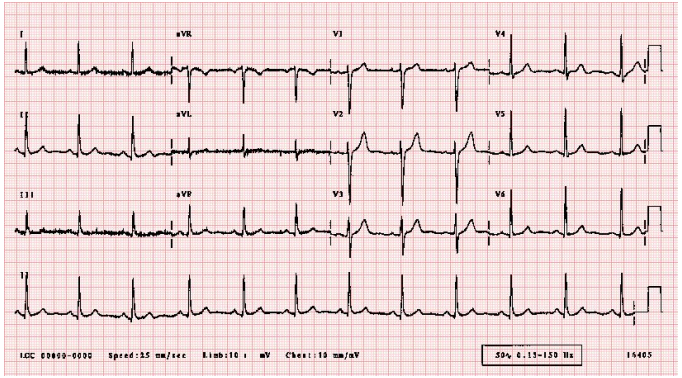
However, this method faces significant challenges: high cost limits accessibility, availability is restricted to specialized facilities, and patients are exposed to radiation during scanning.



Electrocardiogram (ECG) 💗

Electrocardiogram (ECG) shows a line graph of the heart's electrical activity through repeated cardiac cycles. It is an electrogram of the heart which is a graph of voltage over time of the electrical activity of the heart using electrodes placed on the skin.

In a conventional 12-lead ECG, ten electrodes are placed on the patient's limbs and on the surface of the chest. Moreover, it can be used to measure the rate and rhythm of heartbeats, the presence of any damage to the heart's muscle cells or conduction system.



Why ECG predicts Coronary Artery Calcium Score 💗

12 ECG leads can specify to each viewing specific heart regions:

- V1-V4: Anterior/septal wall (Left Anterior Descending: LAD artery)
- II, III, aVF: Inferior wall (Right Coronary Artery: RCA artery)
- I, aVL, V5, V6: Lateral wall (Left Circumflex Artery: LCx artery)

This information detects microscopic electrical abnormalities corresponding to calcified arteries.

The 12-Lead ECG Advantage

The 12-lead ECG is a low-cost, non-invasive, and widely accessiblediagnostic tool available in virtually every clinic worldwide, making it ideal for large-scale cardiac screening.

Hackathon Hypothesis & Goal 💗

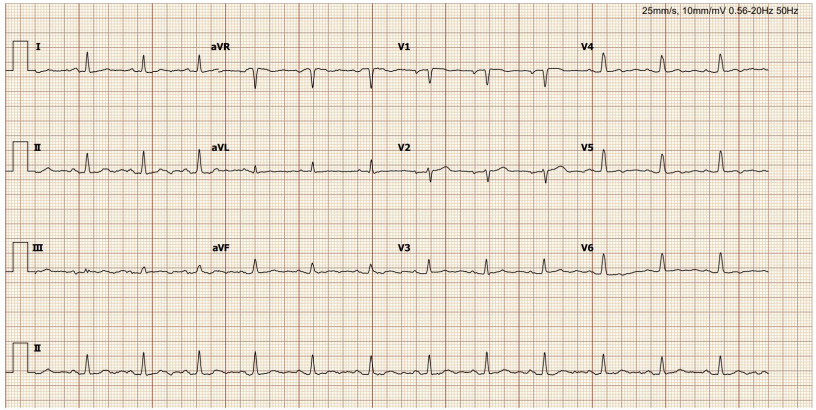
ECG signals contain hidden patterns linked to CAC that are invisible to the human eye. Our goal is to design and develop a deep learning model to predict CAC score (high/low) from a single ECG with CT-level accuracy.

Model Development Pipeline 💗

Data Representation:
Raw waveform, Spectrogram

1D-CNN, 2D-CNN: Resnet,
Transformer-based

Dataset Details 💗



ECG File

1. Image File (.png)
2. Data Point File (.wfdb/.csv)

No	HN	Sex	Age	rca	lma	lad	lcx	ECG	CTA Study d	RCA More	LMA More	LAD More	LCX More	CaScore
1	1065006	หญิง	60	0	0	0	0	Normal	12/4/2013	0	0	0	0	0
2	4778028	หญิง	60	0	0	0	0	abnormal	5/9/2013	0	0	0	0	0
3	4777463	หญิง	62	0	0	0	0	Normal	31/1/2013	0	0	0	0	0
4	1826078	หญิง	45	1	0	0	0	borderline	26/6/2013	1	0	0	0	0
5	4775907	หญิง	55	0	0	0	0	Normal	20/8/2013	0	0	0	0	0
6	4775146	หญิง	53	0	0	0	0	Normal	25/8/2013	0	0	0	0	0
7	4771666	หญิง	59	0	0	0	0	abnormal	17/8/2013	0	0	0	0	0
8	4780592	ชาย	60	0	0	0	0	Normal	18/9/2013	0	0	0	0	0
9	4779783	หญิง	69	0	0	0	0	Normal	30/8/2013	0	0	0	0	0
10	4775146	หญิง	53	0	0	0	0	Normal	25/8/2013	0	0	0	0	0
11	1746458	หญิง	53	0	0	0	0	Normal	20/7/2013	0	0	0	0	0
12	1630423	หญิง	47	0	0	0	0	Normal	28/8/2013	0	0	0	0	0
13	1004756	หญิง	61	0	0	0	0	Normal	6/7/2013	0	0	0	0	0
14	477853	หญิง	59	0	0	0	0	normal	13/12/2011	0	0	0	0	10
15	4783831	ชาย	63	0	0	0	0	Abnormal	15/9/2013	0	0	0	0	56

Patient Infomation File (.csv)

Hand-on Workshop 💗

https://bit.ly/4vqWHoE

BREAKKKKKKKKKK !!!!!!!

Info Session for ASCE Hackathon 💗

Timeline 💗

Final Prediction and Code Submission
Before 18 Apr at 8:00 AM

Workshop & Kickoff

10 Apr 2026

Model Development Phase
10 - 16 Apr 2026

1st Dataset Providing & First Submission on Leaderboard
14 Apr 2026

Hackathon Day 1 - **Final Submission with 2nd Dataset Providing**

17 Apr 2026

Hackathon Day 2 - **Presentation Day**

18 Apr 2026

Submission Process💗

It is required to submit your prediction csv file, code, and slide before 18 Apr at 8:00

Production CSV Submission Process

Dataset → Model → Prediction (CSV format) —Submit via your page→ Leaderboard (first time on 14 Apr)

Code & Slide PDF Submission Process

via github classroom

Join the classroom → Create/Join the group → Accept the assignment

Submit in github

Presentation💗

10 teams will have to do presentation for 5 minutes and Q&A session for 3 minutes per team

Award Criteria

1. Model Performance (Score Metrics)
2. Model Training Experiment and Approaches
3. Presentation Clarity and Understanding
4. Visualization and Model Application

Q&A Session